

Q.5

Solⁿ

$$\alpha = (1, 3, 0) \quad \beta = (2, 1, -2)$$

$$\begin{aligned} L\{\alpha, \beta\} &= c_1(1, 3, 0) + c_2(2, 1, -2) \\ &= (c_1 + 2c_2, 3c_1 + c_2, -2c_2). \end{aligned}$$

$$\gamma = (-1, 3, 2) = (c_1 + 2c_2, 3c_1 + c_2, -2c_2)$$

$$\Rightarrow \underbrace{c_1 + 2c_2 = -1}_{\Rightarrow c_1 = 1}, \quad 3c_1 + c_2 = 3, \quad -2c_2 = 2 \Rightarrow \boxed{c_2 = -1}$$

$$\text{Now } 3c_1 + c_2 = 3 \text{ for } c_1 = 1, c_2 = -1$$

$$\Rightarrow 3 - 1 = 3 \text{ (Not true)}$$

$$\Rightarrow \gamma \notin L\{\alpha, \beta\}.$$

Similarly, we can find for δ .

Q.6

$$\alpha_1, \alpha_2, \alpha_3 \in V \text{ such that}$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 0. \quad \text{--- ①}$$

Now,

$$L\{\alpha_1, \alpha_2\} = L\{\alpha_1, \alpha_3\} \text{ (To prove).}$$

Proof:-

$$\text{Let } u \in L\{\alpha_1, \alpha_2\}$$

$$\Rightarrow u = c_1\alpha_1 + c_2\alpha_2$$

$$\Rightarrow u = c_1(-\alpha_2 - \alpha_3) + c_2\alpha_2 \quad [\because \text{from ①}]$$

$$= (-c_1 + c_2)\alpha_2 - c_1\alpha_3$$

$$\Rightarrow u \in L\{\alpha_2, \alpha_3\}$$

Similarly, we can prove for the rest.

Q.7

Solⁿ

$$\alpha_1 = (1, 2, 0), \quad \alpha_2 = (3, -1, 1), \quad \alpha_3 = (4, 1, 1) \quad \text{Let } S = \{\alpha_1, \alpha_2, \alpha_3\}$$

Let

$$c_1(1, 2, 0) + c_2(3, -1, 1) + c_3(4, 1, 1) = (0, 0, 0)$$

$$\Rightarrow c_1 + 3c_2 + 4c_3 = 0$$

$$2c_1 - c_2 + c_3 = 0$$

$$c_2 + c_3 = 0$$

$$\Rightarrow \boxed{c_2 = -c_3}$$

$$2c_1 = -2c_3$$

$$\Rightarrow \boxed{c_1 = -c_3}$$

$$\Rightarrow \boxed{c_1 = c_2 = -c_3}, \Rightarrow \text{L.D.}$$

(Since you can take $c_1=1, c_2=1, c_3=-1$).

$$L(S) = c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3$$

$$\text{Now, } \alpha_3 = \alpha_1 + \alpha_2$$

$$\Rightarrow L\{\alpha_1, \alpha_2, \alpha_3\} = L\{\alpha_1, \alpha_2\}.$$

$S' = \{\alpha_1, \alpha_2\}$ is a proper subset of S that generate $L(S)$.

Q.8

Solⁿ

$$\text{① True as } 1 \cdot 0 = 0 \quad [\Rightarrow \text{L.D.}]$$

$$\text{②. False } \underbrace{\{(1, 1, 0), (2, 2, 0)\}}_{\text{L.D.}} \subset \mathbb{R}^3$$

$$\text{but } (1, 1, 0) \rightarrow \text{L.I.}$$

$$\text{③. Let } S' \subseteq S, \text{ where } S \text{ is L.I.}$$

To show

$$\text{if } c_1v_1 + c_2v_2 + \dots + c_nv_n = 0, \quad v_i \in S'$$

$$\Rightarrow c_i = 0, \quad \text{for all } i.$$

$$\text{Since, } S' \subseteq S$$

$$\Rightarrow v_i \in S, \quad \text{but } S \text{ is L.I.}$$

$$\Rightarrow c_i = 0.$$

H. P.